**Answer ALL questions. ALL working should be shown.**

1.

(a) Given:

$$\hat{X}_A = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \hat{Y}_A = \begin{bmatrix} 0.2588 \\ 0.9659 \\ 0 \end{bmatrix}, \hat{Z}_A = \begin{bmatrix} 0.9659 \\ -0.2588 \\ 0 \end{bmatrix}$$

$$\hat{X}_B = \begin{bmatrix} 0.3536 \\ 0.6124 \\ -0.7071 \end{bmatrix}, \hat{Y}_B = \begin{bmatrix} -0.1268 \\ 0.7803 \\ 0.6124 \end{bmatrix}, \hat{Z}_B = \begin{bmatrix} 0.9268 \\ -0.1268 \\ 0.3536 \end{bmatrix}$$

Calculate the rotation matrix ${}^A_BR$. You must show the steps, not just the final answer.

[5 marks]

(b) An isosceles hexagon has equal edge lengths of 1, as shown in Figure 1.1:

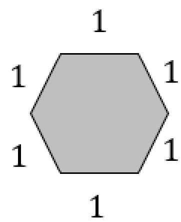


Figure 1.1: Isosceles hexagon

The centre of the hexagon is at $(1,3)$ with respect to the robot base frame. One edge of the hexagon is initially parallel to the $x$-axis of the robot. The robot then rotated the hexagon about $(1,3)$ by $45°$, as shown in Figure 1.2:

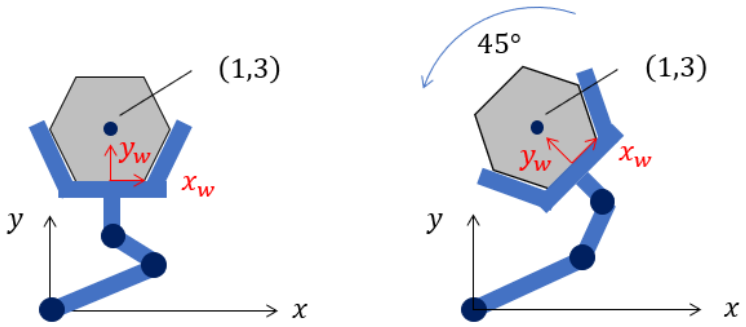


Figure 1.2: Hexagon rotated by robot by $45°$

The wrist frame is denoted by $x_w$ and $y_w$ axes. What is the position of the wrist frame after the $45°$ rotation, as shown in the right figure in Figure 1.2?

[5 marks]

(c) Where is the centre of the hexagon with respect to the wrist frame, before and after the rotation shown in Figure 1.2 above?

[1 mark]



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(d) Given:

$$ {}^{A}_{B}T = \begin{bmatrix} 0.3536 & 0.1768 & 0.9186 & 2 \\ 0.3536 & 0.8839 & -0.3062 & 5 \\ -0.8660 & 0.4330 & 0.2500 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Calculate ${}^{A}_{B}T^{-1}$. You must show the steps, not just the final answer.

[2 marks]

(e) Frames $\{A\}$ and $\{B\}$ are initially coincident. First, rotate $\{B\}$ about $Y_B$ by $30°$. Next, rotate $\{B\}$ about $Z_B$ by $45°$. Finally, rotate $\{B\}$ about $Y_B$ by $45°$.

Calculate the rotation matrix ${}^{A}_{B}R$, then use your answer to calculate the Euler parameters (Quaternions). You must show the steps, not just the final answer.

[6 Marks]

2. A 2-DOF RP robot is placed on a linear stage, as shown in Figure 2.1.

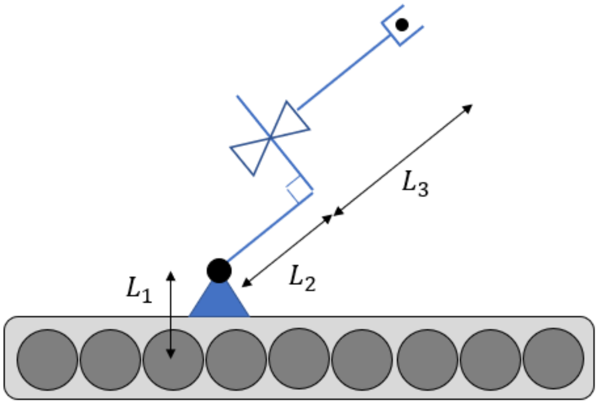


Figure 2.1: A 2-DOF RP robot placed on a linear stage

(a) Derive the forward kinematics model of the robot (including the linear stage), and show that the position of the end-effector is:

$$^{0}P = \begin{bmatrix} L_3c_2 + L_2c_2 - d_3s_2 + L_1 \\ 0 \\ -L_3s_2 - L_2s_2 - d_3c_2 + d_1 \\ 1 \end{bmatrix}$$

Note:

- If there is any ambiguity in up-down direction, choose up.
- If there is any ambiguity in left-right direction, choose right.
- If there is any ambiguity in into the plane / out of the plane, choose out of the plane.
- If there is any ambiguity in the diagonal direction, choose left+up.

[10 marks]

(b) Derive the linear Jacobian matrix of the end-effector, with respect to the robot's frame {0}. Then calculate the joint torque / force is an external force of $^{0}F = [-1 \quad 0 \quad -1]^T$ acts on the end-effector when $d_1 = 1\ \mathrm{m}$, $\theta_2 = -90°$, $d_3 = 0\ \mathrm{m}$. Include the unit for each component.

[4 marks]

3.

(a) Given an RP robot as shown in Figure 3.1.

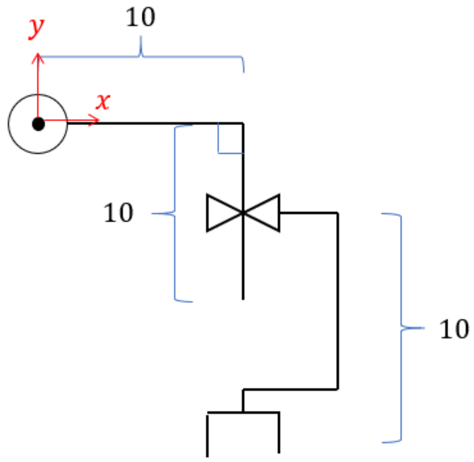


Figure 3.1: RP robot

Sketch the workspace of the robot. Please include dimensions of the workspace.

If the end-effector needs to move from $(15,10)$ to $(5,-17)$, would you plan a trajectory in joint space or Cartesian space? Justify your choice.

[3 marks]

(b) To further reduce the wear and tear of the robot, you decided to increase the order of the trajectory polynomial to seven (instead of five in the case of Quintic polynomial), i.e.

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 + a_6 t^6 + a_7 t^7$$

Calculate the parameters $a_i$ such that:

$$x(0) = u_0$$
$$x(t_f) = u_f$$
$$\dot{x}(0) = \dot{x}(t_f) = 0$$
$$\ddot{x}(0) = \ddot{x}(t_f) = 0$$
$$\dddot{x}(0) = \dddot{x}(t_f) = 0$$

[14 marks]

4. An RP robot is shown in Figure 4.1.

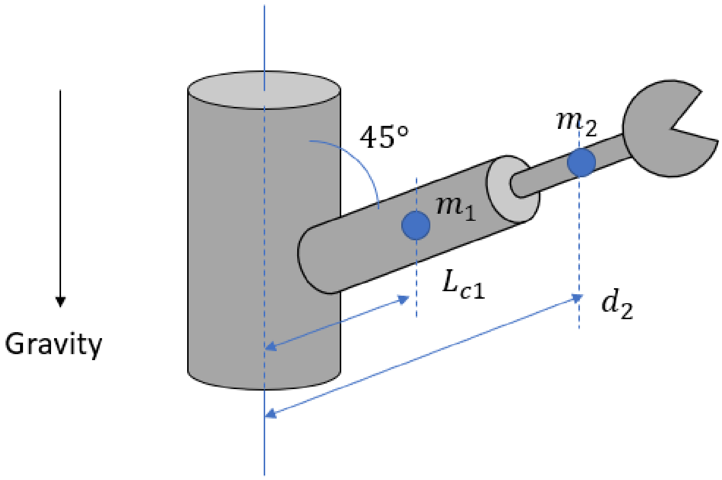


Figure 4.1: RP Robot

$m_1$ and $m_2$ are the masses of each link, and the blue dots represent the centres of mass. The origin of frame {2} is located at the same position as centre of mass of the second link (i.e. directly on the blue dot labelled $m_2$). Also, the inertia tensor of the two links with respect to their centres of mass are:

$$^{C1}I_1 = \begin{bmatrix} I_{XX1} & 0 & 0 \\ 0 & I_{YY1} & 0 \\ 0 & 0 & I_{ZZ1} \end{bmatrix}$$

$$^{C2}I_2 = \begin{bmatrix} I_{XX2} & 0 & 0 \\ 0 & I_{YY2} & 0 \\ 0 & 0 & I_{ZZ2} \end{bmatrix}$$

Derive the dynamic equations of the robot. Write the dynamic equations in the $M(q)\ddot{q} + V(q,\dot{q}) + G(q) = \tau$ form.

[25 Marks]

5.

The transformation matrices for an ABB IRB120 robot are as follows:

$$ {}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^2_3T = \begin{bmatrix} c_3 & -s_3 & 0 & a_2 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^3_4T = \begin{bmatrix} c_4 & -s_4 & 0 & a_3 \\ 0 & 0 & 1 & d_4 \\ -s_4 & -c_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^4_5T = \begin{bmatrix} c_5 & -s_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_5 & c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^5_6T = \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

You are also given the following matrix-inverses:

$$ {}^0_2T^{-1} = \begin{bmatrix} c_1c_2 & s_1c_2 & -s_2 & 0 \\ -c_1s_2 & -s_1s_2 & -c_2 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0_3T^{-1} = \begin{bmatrix} c_1c_{23} & s_1c_{23} & -s_{23} & -a_2c_3 \\ -c_1s_{23} & -s_1s_{23} & -c_{23} & a_2s_3 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}_4^0T^{-1} = \begin{bmatrix} c_1c_{23}c_4 + s_1s_4 & s_1c_{23}c_4 - c_1s_4 & -s_{23}c_4 & * \\ -c_1c_{23}s_4 + s_1c_4 & -s_1c_{23}s_4 - c_1c_4 & s_{23}s_4 & * \\ -c_1s_{23} & -s_1s_{23} & -c_{23} & * \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}_5^0T^{-1} = \begin{bmatrix} c_1c_{23}c_4c_5 + s_1s_4c_5 - c_1s_{23}s_5 & s_1c_{23}c_4c_5 - c_1s_4c_5 - s_1s_{23}s_5 & -s_{23}c_4c_5 - c_{23}s_5 & * \\ -c_1c_{23}c_4c_5 - s_1s_4s_5 - c_1s_{23}c_5 & -s_1c_{23}c_4c_5 + c_1s_4s_5 - s_1s_{23}c_5 & s_{23}c_4s_5 - c_{23}c_5 & * \\ c_1c_{23}s_4 - s_1c_4 & s_1c_{23}s_4 + c_1c_4 & -s_{23}s_4 & * \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

where * in ${}_4^0T^{-1}$ and ${}_5^0T^{-1}$ means “don’t care”.

The 6$^{th}$ frame of the robot is required to be at:

$$ {}^0P_{6ORG} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} $$

And its orientation is required to be:

$$ \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

Calculate the joint angles to achieve the desired position and orientation of the 6$^{th}$ frame.

[25 marks]



**END OF PAPER**